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# Build A Universal AC/DC Booster, Audio Amplifier Cum Line Driver

A universal AC/DC booster, audio amplifier and line driver can be used in sensor circuits, audio signal generators, electrical guitars and so on. Most of these applications need high input impedance (for example, 10-mega-ohm), low quiescent current, output current limiting function, low total harmonic distortion (THD) and intermodulation distortion (IMD), and are capable of driving loads as low as 32-ohm, like headphones and high-impedance loudspeakers.

## PARTS LIST

### Semiconductors:

- IC1 - TL072 op-amp
- T1 - BD135 npn transistor
- T2 - BD136 pnp transistor
- D1, D2 - 1N4148 signal diode

### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R15 - 1-kilo-ohm
- R2 - 10-mega-ohm
- R3 - 18-kilo-ohm
- R4 - 2-kilo-ohm
- R5, R7 - 10-kilo-ohm
- R6 - 4.7-kilo-ohm
- R8, R9 - 6.8-kilo-ohm
- R10, R11 - 56-ohm
- R12, R13 - 2-ohm
- R14 - 47-ohm

### Capacitors:

- C1-C3 - 0.1 $\mu$ F ceramic disk
- C4, C5 - 220 $\mu$ F, 63V electrolytic
- C6 - 33nF ceramic disk
- C7 - 10pF ceramic disk

### Miscellaneous:

- CON1, CON3 - 3-pin connector
- CON2 - 2-pin connector
- 8-pin IC base
- Heat-sink for BD135 and BD136

(All these are available from kitsnspares.com)

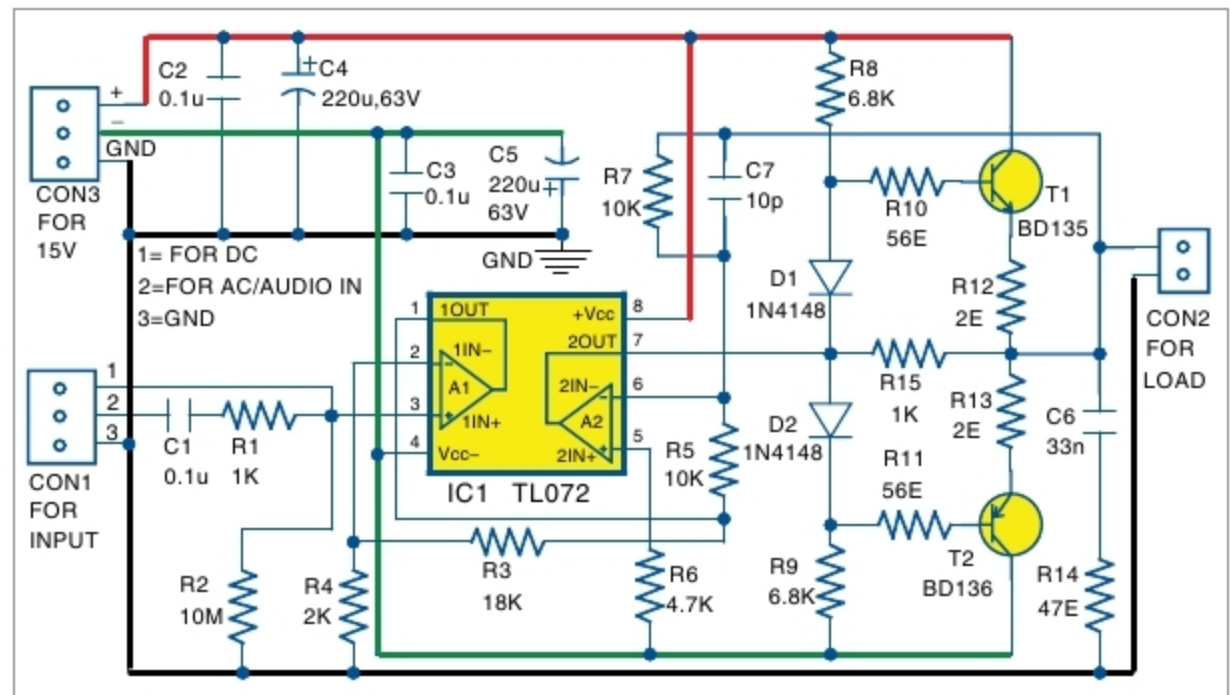


Fig. 1: Circuit diagram of the universal AC/DC booster, audio amplifier and line driver

This circuit produces  $\pm 300\text{mA}$  output current and  $20\text{Vpp}$  ( $\pm 10\text{V}$ ) output voltage with power supply of  $\pm 15\text{V}$ . It is simple and useful for a large variety of applications.

## Circuit and working

The circuit diagram of the universal circuit is shown in Fig. 1. It is built around op-amp TL072 (IC1), transistor BD135 (T1), transistor BD136 (T2), diodes 1N4148 (D1 and D2) and a few other components.

The circuit has two operational amplifier stages (A1 and A2). Both stages are built around dual JFET operational amplifier TL072 (IC1). You may replace IC1 with TL082, LF412, LF353, etc.

You can apply AC or DC, or audio input signal to connector CON1.

Capacitor C1 is used if the circuit is used as AC booster or audio amplifier. Input resistance is determined by resistor R2 (10-mega-ohm), which can be set to any appropriate value.

Op-amp A1 of IC1 operates as DC amplifier whose gain is determined by the following relationship:

$$A_{v1} = 1 + R3/R4 = 10$$

You may use any appropriate gain that is  $\geq 1$

Op-amp A2 of IC1 operates as DC amplifier whose gain is determined by the following relationship:

$$A_{v2} = -R7/R5 = -1$$

Total gain  $A_v$  of the circuit is  $A_{v1} \times A_{v2} = 10 \times (-1) = -10$

Output stage is built around two bipolar transistors (T1 and T2). R8, D1, D2, R9 and R15 are used to reduce THD and IMD of the amplifier. Resis-

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